

ROBERT M. BERNSTEIN, MD, FACE
Regional Endocrinology Associates, PC
1580 Cerro Gordo Road
Santa Fe, NM 87501
Office/Cell: 505-501-0200
Email: robertbernsteinmd@comcast.net

October 28, 2023

Rachel Raymond
1151 El Centro Street, Suite D
South Pasadena, CA 91030
818.395.4750

Re: Athena Castaneda (DOB: 09/21/2021) and Elleonore Castaneda (DOB: 06/17/2020)

Dear Ms. Raymond:

You have asked me to review Athena and Elleonore Castaneda's medical records, including genetic testing, and to give my opinion regarding Athena and Elleonore's multiple fractures.

QUALIFICATIONS

I am a Board-Certified Endocrinologist with over 45 years of experience who was in private practice in Santa Fe, New Mexico from August 1979 until May 2016, at which time I became a Consulting Endocrinologist at La Familia Medical Center, a government-funded low-income clinic in Santa Fe, where I worked until November 2021. Since that time I have been a consulting Endocrinologist at Santo Domingo Pueblo Health Center, which serves the Native American community of Santo Domingo Pueblo. In the past, as outlined on my CV, I have also acted as a Consulting Endocrinologist for Presbyterian Health Services in Albuquerque, New Mexico; Cancer Treatment Centers of America in Newnan, Georgia; and Navicent Health Center in Macon, Georgia.

I earned my medical degree from Case Western Reserve University School of Medicine (Cleveland, Ohio) in 1970, completed my Internship and Residency in Internal Medicine at Boston City Hospital from 1970-1975 (Boston, Massachusetts), and my Fellowship in Endocrinology and Metabolism at Stanford University Medical Center (Palo Alto, California) from 1975-1977.

I am a member of the following professional organizations: American Diabetes Association; Endocrine Society; American Association of Clinical Endocrinologists.

As a clinical endocrinologist who treats patients with metabolic and genetic bone diseases, I am knowledgeable regarding the medical literature related to these conditions, the therapies available, and standards of care.

RECORDS AND DOCUMENTS REVIEWED

Athena Castaneda Medical Records Reviewed: MemorialCare Long Beach, MemorialCare Medical Group, Loma Linda University Children's Hospital, Providence St. Mary Medical Center, Children's Hospital of Orange County, Predictiv Genetics Report

Elleonore Castaneda Medical Records Reviewed: Dr. Teresita Salazar, MemorialCare Long Beach, MemorialCare Medical Group, Loma Linda University Children's Hospital, Providence St. Mary Medical Center, Children's Hospital of Orange County, Genetics Center, Invitae Genetics Report, Predictiv Genetics Report

Claudia Sandoval Medical Records Reviewed: MemorialCare, Providence, Invitae Genetics Report

Physician Opinions

County of San Bernardino CFS Case Social Worker's Reports

SUMMARY OF MEDICAL HISTORY

Elleonore Castaneda was born to mother, Claudia Sandoval, and father, Nick Castaneda, at St. Mary Medical Center in Long Beach, CA on 6/17/2020. Elleonore had to be manipulated by the attending physician in order to achieve vaginal birth. Significant findings at birth included jaundice, decreased platelets, and ABO incompatibility.

Athena Castaneda was born to mother, Claudia Sandoval, and father, Nick Castaneda, at Providence St. Mary Medical Center in Apple Valley, CA on 9/25/2021. Athena had to be manipulated by the attending physician in order to achieve vaginal birth. Significant findings at birth included ABO incompatibility, abnormal hematologic findings (see below) and placental abruption. Also, it is noted that Athena didn't pass the hearing test until the fourth attempt.

It is also notable that the mother, Claudia, suffered multiple infections and severe anemia with all pregnancies and successfully accessed appropriate prenatal care for all children.

Patient Athena Castaneda presented with bilateral parietal bone fractures and a 2 mm subdural hematoma and overlying scalp contusion at Loma Linda University Emergency Department on 10/15/2021 just weeks after birth. Athena was subsequently admitted and further testing was completed. Imaging reports display a range of bone abnormalities in Athena's ribs and legs; however, the rib fractures were determined to be normal variants (no fractures were found). Athena was treated for Vitamin D deficiency while admitted at LLUCH and was discharged with a prescription for Vitamin D to continue treatment at home. Athena's Glasgow Coma Scale (GCS) score for her suspected head trauma was documented at 15. The Head Injury Severity Scale (HISS) indicates the severity as follows: minor head injury- to no injuries indicated (HI) (GCS score = 13-15), moderate HI (GCS score = 9-12) and severe HI (GCS score = 3-8).¹

On 10/16/2021, Athena's sister, Elleonore, was admitted at LLUCH to complete a non-accidental trauma evaluation. Imaging showed a healed skull fracture and two rib abnormalities *thought* to be healing fractures. Elleonore was diagnosed with high anion gap metabolic acidosis and was under the care of a gastroenterologist. Both children were discharged from LLUCH on 10/20/2021.

A referral was made for a genetic evaluation for both Athena and Elleonore. Further testing was completed and it was confirmed that Elleonore had a TNFRSF11A mutation. With the relevant clinical history considered, and abnormal endocrine findings, genetics felt it was necessary for Elleonore and Athena to be seen by a metabolic bone specialist/endocrinologist to rule out congenital bone dysplasias and an organic cause for the bone abnormalities seen.

Athena was admitted to MemorialCare in February of 2022, at which time Athena suffered a near-fatal event due to a severe spontaneous upper GI bleed and sepsis. Athena slipped into respiratory failure but survived. Per MemorialCare records, Athena and Elleonore have both suffered multiple infections between October of 2021 to the present time, for which treatment needed to be sought. Infections documented in the children's records included but are not limited to: COVID, cellulitis (skin infection), adenovirus, rhino/enterovirus, moraxella, norovirus, streptococcus pneumoniae and other upper respiratory infections of unknown origin. Further testing showed immune system dysfunction in both children. A follow up at the Genetics Center for Elleonore Castaneda was performed; however, Dr. Angela Myers could not confirm a diagnosis since she was not provided a full medical history or previous relevant medical records.

DISCUSSION AND OPINIONS

Both Athena and her sister Eleonore have a severe congenital abnormality due to a mutation in a gene named TNFRSF11A. If this gene is normal (i.e., not mutated), it provides appropriate instructions for making a protein that plays an important role in bone formation. Children who have an abnormal version (a mutation) of this gene will develop abnormal bones along with osteolytic lesions due to constant bone remodeling.² In the case of Athena and Elleonore, this congenital abnormality has caused familial expansile osteolysis, a metabolic bone disorder in which their bones are extremely fragile, and often fracture with minimal pressure.² Familial expansile osteolysis or FEO is an autosomal dominant disorder.² This means that only one parent needs to have a copy of the mutated gene for the mutation to be disease causing. In this case, the mother, Claudia Sandoval, is the confirmed genetic carrier.

In confirming that a mutation is pathogenic (disease causing), one must also assess if correlating clinical abnormalities are present. Patients with metabolic bone disorders present with abnormal endocrine laboratory findings in several of the following components: alkaline phosphatase, calcium, Vitamin D, PTH, magnesium, phosphate and albumin.^{3,4,5,6}

Athena presented at Loma Linda University Children's Hospital (LLUCH) with significantly increased alkaline phosphatase, decreased levels of Vitamin D,⁴ elevated AST, decreased albumin and elevated phosphorus. Although Athena's calcium level was presented as normal, an adjusted calcium needed to be obtained due to the decreased albumin level.⁷ This was not completed at LLUCH. Athena also had decreased serum total protein, which is another marker utilized in the diagnosis of bone marrow and metabolic disorders.⁸ Athena was administered Vitamin D while admitted at LLUCH and received a prescription for Vitamin D to continue treatment at home. It should also be noted that Elleonore was prescribed Vitamin D by Dr. Teresita Salazar on 8/28/2020, showing that Elleonore had been receiving appropriate supplementation.

Athena and Elleonore not suffering new fractures only supports their diagnosis since the appropriate therapy was utilized and successful. Also, Athena was put on enhanced baby

formula fortified with Vitamin D and calcium, which was also beneficial. It should be noted that the children will need repeat imaging on a regular basis to determine if there are new lesions, fractures and/or bone deformities since fractures are only one component of this disease. They will also need periodic laboratory testing, likely at a minimum of twice a year, and probably more frequently.

Elleonore presented at LLUCH with significantly increased alkaline phosphatase, elevated calcium, elevated AST, elevated CK (elevated CK, or creatine kinase, is used as a biomarker for genetic osteopetrosis⁹) and high anion gap metabolic acidosis. No levels of Vitamin D, PTH or magnesium were obtained.

Elleonore also presented at LLUCH with a healed skull fracture. Per a thorough multi-specialist workup at MemorialCare, the skull fracture was determined to be most likely an injury from birth. It should be noted that several generations of this family, including the maternal grandfather, suffered from confirmed skull and bone abnormalities at birth.

As noted above, both Athena and Eleonore have a severe congenital abnormality due to a mutation in a gene named TNFRSF11A. Children with this problem, in addition to fragile bones, can also have hematologic abnormalities (disorders of the blood) as both Athena and Eleonore have. In Athena and Eleonore's case, the platelet abnormalities are especially significant, since platelets are critical for the ability of blood to clot.

At birth, at Providence St. Mary, Athena was found to have ABO Incompatibility, DAT IgG positive, decreased MCV (red blood cells smaller than normal), as well as other abnormal hematologic findings.

At LLUCH, Athena presented with decreased red blood count (RBC), decreased hemoglobin (the molecule that carries oxygen within the red blood cells), and significantly elevated platelets, among other hematologic abnormalities.

While admitted at MemorialCare Long Beach during February 2022, Athena again presented with the same hematologic abnormalities seen at LLUCH, such as significantly elevated platelets, decreased hemoglobin, decreased red blood count and decreased hematocrit. Athena also had an elevated PTT and abnormal mixing study, indicative of a bleeding disorder.¹¹ The hospitalization occurred due to a spontaneous upper GI bleed and hematemesis (vomiting blood). Athena was also septic and went into respiratory failure. After Athena was discharged, MemorialCare completed further hematologic studies. Athena was diagnosed with thrombocytosis (excess platelets) and congenital Von Willebrand's disease. These are both rare bleeding disorders that cause both abnormal bleeding and clotting. This means that any bruising or ruptured vessels could happen unprovoked or due to minimal pressure.

At birth, Eleonore presented at Providence St. Mary with ABO incompatibility, DAT positive, elevated hemoglobin, and decreased platelets.

At LLUCH, Eleonore presented with blood in her urine and an elevated prothrombin time. The elevated prothrombin time indicates that Eleonore's blood is not able to clot within an appropriate amount of time, and therefore Eleonore is subject to bruises and bleeding that may occur spontaneously or with minor trauma. This finding of an elevated prothrombin time, along with recorded platelet abnormalities from Eleonore's birth records, should have necessitated further hematologic inquiry to identify the root cause of Eleonore's abnormal bleeding.

As far as the forensic reports, standard of care dictates that when a non-accidental trauma is suspected, the first diseases that should be ruled out via differential diagnosis are metabolic bone disorders and bleeding disorders, which both children unfortunately have. ^{12,13,14,15,16,17,18,19}

Addendum: On 10/25/2023, I was given updated medical records from Loma Linda regarding Athena's current health. These records noted that Athena has seen both a geneticist and endocrinologist. However, the last genetics report for Athena by Dr. Zadeh at CHOC also stated that Elleonore needed to be fully assessed by an endocrinologist/metabolic bone specialist. When Elleonore's testing was completed, further evaluation was to be done for Athena. I see multiple referrals to endocrinology, but not any actual encounters for either child. Also, many of Athena's diagnoses are missing from the records, including Von Willebrand's Disease and thrombocytosis. The newest record also does not mention the followup referrals to hematology and immunology, which I consider extremely important, since both Athena and Elleonore have suffered multiple infections and have abnormal lab findings consistent with immune system dysfunction.

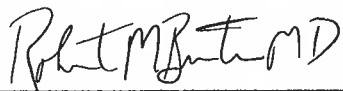
SUMMARY

In summary, based on Athena's and Elleonore's clinical histories as well as the congenitally abnormal pathologies and genetic analysis, it is my opinion that, more probable than not, that both Athena's and Elleonore's fractures and bruising were caused by the congenital abnormalities suffered by both children, rather than from abuse.

Furthermore, based on medical necessity, my opinion is that both Athena and Elleonore need immediate and ongoing medical care in order to prevent lifelong bone deformities and consequent disabilities, and that this medical care must be directed by physicians who are expert in the care of children who suffer from the conditions with which Athena and Elleonore have been diagnosed. These experts must include an Endocrinologist well-versed in metabolic bone disease, and an Orthopedist.

If you need any further information, please do not hesitate to contact me. The opinions expressed above are subject to modification if further information becomes available and are expressed to a reasonable degree of medical probability.

Respectfully submitted,


Robert M. Bernstein, M.D.

Date: October 28, 2023

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CURRICULUM VITAE

Robert M. Bernstein, MD, FACE
Regional Endocrinology Associates, PC
1580 Cerro Gordo Road, Santa Fe, New Mexico, 87501
Cell Phone: (505) 501-0200
Email: robertbernsteinmd@comcast.net

COLLEGE: Case Western Reserve University
Cleveland, Ohio; Bachelor of Science, 1967

MEDICAL: Case Western Reserve University School of Medicine
Cleveland, Ohio; MD, 1970

INTERNSHIP, Internal Medicine: Boston City Hospital, 1970-71

RESIDENCY, Internal Medicine: Boston City Hospital, 1971-1972, 1973-1974

FELLOWSHIP, Endocrinology and Metabolism:
Stanford University Medical Center, 1975 - 1977

CLINICAL PRACTICE:
Internal Medicine, Incline Village, NV, 1977-79
Regional Endocrinology Associates, PC,
Santa Fe, NM, August 1979 to May 2016
Consulting Physician, Endocrinology and Diabetes,
La Familia Medical Center (government-funded low-income clinic)
Santa Fe, NM, October 2015 - December 2021
Consulting Physician, Endocrinology and Diabetes,
Santo Domingo Pueblo Health Center,
Santo Domingo, NM, October 2021- Current

CERTIFICATION: American Board of Internal Medicine, 1975
Endocrinology and Metabolism, 1977

LICENSES: New Mexico, California

PUBLICATIONS:

Non-Ketotic Diabetes Mellitus: Insulin Deficiency or Insulin Resistance?
American Journal of Medicine 60:80, 1976 (coauthor)

Hepatic Insulin Responsiveness in Patients with Endogenous Hypertriglyceridemia. Diabetologia 14:249, 1978
(coauthor)

Effect of Obesity on the Relationship Between Very Low Density Lipoprotein Production Rate and Plasma Triglyceride Concentration in Normal and Hyperglycemic Subjects. Metabolism 27:1047, September, 1978 (coauthor)

Defect in Glucose Removal in Non-Ketotic Diabetic Patients with Fasting Hyperglycemia. Diabetes 28:32, 1979 (coauthor)

AACE Clinical Practice Guidelines for the Management of Thyroid Carcinoma. Endocrine Practice 3:62, 1997 (coauthor)

Use of an Automated Device for Alternative Site Blood Glucose Monitoring. Diabetes Care 24:1217, July 2004 (coauthor)

Evaluation of a Simple, Easy-to Use Blood Glucose Meter (Precision Xceed), Abstract, American Diabetes Association Scientific Meeting, 2004 (coauthor)

Evaluation of the FreeStyle Lite Blood Glucose Monitoring System, Internal Publication, Abbott Diabetes Care, 2007 (coauthor)

Criteria for Continuous Glucose Monitoring Systems. Presbyterian HealthCare Systems Policy Statement, October, 2007 (coauthor)

Benefits of Insulin Pump Therapy, and Problems. In Insulin Pump Safety Meeting: Summary Report, by Klonoff and Reyes, Journal of Diabetes Science and Technology 3(2): 396, March 2009 (contributor)

Performance of a New Test Strip for FreeStyle Blood Glucose Monitoring Systems. Diabetes Technology and Therapeutics 13(1):1, 2011 (coauthor)

Accuracy and Precision of Aviva Freedom 2 Glucose Meter, Abstract, Advanced Technologies and Treatments for Diabetes Conference, Barcelona, Spain, February 2012 (coauthor)

User Performance Evaluation Of The Contour® Next Blood Glucose Monitoring System With The Contour® Next Test Strip, Poster, Diabetes Technology Society Meeting, Bethesda, MD, November 2012 (coauthor)

Accuracy, Precision, and User Performance Evaluation Of The Contour® Next Blood Glucose Monitoring System With The Contour® Next Test Strip, 6th International Conference on Advanced Technologies & Treatments for Diabetes, Paris, France, February 2013 (coauthor)

Multicenter Evaluation of the ACCU-CHEK Meters (Inform 2, Aviva, and Performa) with High Performance Test Strips: Capillary Blood Testing (coauthor)

EDITORIAL REVIEW BOARD: Endocrine Research, 2009-present

TEACHING EXPERIENCE:

Clinical Associate, University of New Mexico 1979-1988:
Supervision and teaching of Internal Medicine residents

Supervising Physician, Endocrinology and Diabetes:
Residents rotating through Christus St. Vincent Hospital Family
Practice Program at La Familia Medical Center, 2015 - 2021

Numerous lectures and teaching discussions with
physicians, nurses, physicians in training, and
lay groups. Extensive public speaking experience.

MEDICAL-LEGAL: Have reviewed approximately 40 medical records, both for plaintiffs and defendants, from 1995 to present, for quality and appropriateness of care. 17 depositions and nine court appearances as an expert witness.

CLINICAL STUDIES: Glucose Meter Studies:
Approximately 150 clinical evaluations of various glucose meters for accuracy and precision, ease of use, effect of altitude, approximately 8500 patients, 1991-present.

Other Studies:

- Intensive evaluation of continuous glucose sensor, with venous blood sampling every 15 minutes for 72 hours.
- Evaluation of an instrument for point-of-service urine microalbumin.
- Evaluation of an instrument for point-of-service HgbA1c.
- Evaluation of outpatient use of continuous glucose monitoring systems.
- Professional evaluation of various diabetes software.

PROFESSIONAL SOCIETIES: American Diabetes Association
American Association of Clinical Endocrinologists
Endocrine Society

CERTIFICATION: Certified Diabetes Technology Clinician (Diabetes Technology Society)

LECTURES
(1977-Present)

Pathogenesis and Treatment of Hyperlipidemia, Entire medical staff, Santa Clara Valley Medical Center, San Jose, CA, 1977.

Alcoholic Ketoacidosis, Entire medical staff, Santa Clara Valley Medical Center, 1977.

Vitamin D Metabolism (grand rounds), Entire medical staff, Santa Clara Valley Medical Center, 1977.

Hyperprolactinemia, Internal Medicine Section, St. Vincent Hospital, Santa Fe, NM, 1980.

Evaluation of Amenorrhea, Entire medical staff, St. Vincent Hospital, Santa Fe, NM, 1980.

Thyroid Carcinoma, Entire medical staff, St. Vincent Hospital, 1980.

Update in Diabetes, Internal Medicine Section, St. Vincent Hospital, 1980.

Hyperprolactinemia, Entire medical staff, St. Vincent Hospital, 1980.

Appropriate Use of Oral Hypoglycemic Agents, Internal Medicine Section, St. Vincent Hospital, 1980.

Update on Oral Agents, Internal Medicine Section, St. Vincent Hospital, 1980.

Recognition and Management of Diabetic Emergencies: Hypoglycemia and Ketoacidosis, Intensive care unit nurses, St. Vincent Hospital, 1981.

Recognition and Management of Endocrine Emergencies, Symposium on Critical Care, San Juan Regional Medical Center, Farmington, New Mexico, 1983.

Recognition and Management of Diabetic Emergencies: Hypoglycemia and Ketoacidosis, Entire nursing staff, St. Vincent Hospital.

Management of Diabetes, Entire medical staff, Los Alamos Medical Center, Los Alamos, NM.

Management of Type 2 Diabetes, State Meeting of American Association of Family Practice, Albuquerque, NM, 1985.

What is Diabetes, Really?, Clinical Diabetes Update, statewide symposium on diabetes, Santa Fe, NM, 1985.

Can Diabetics Eat Ice Cream? (Dietary therapy in diabetes), Clinical Diabetes Update, statewide symposium on diabetes, Santa Fe, NM, 1985.

Diabetes Update, Entire medical staff, Mercy Medical Center, Des Moines, Iowa, 1985

Relative Prevalence of Renal Failure in Type 1 vs. Type 2 Diabetes (Poster Presentation), New Mexico Society of Internal Medicine, New Mexico Scientific Meeting, Albuquerque, NM.

Appropriate Care of Type 2 Diabetes, Entire medical staff, Miners' Colfax Medical Center, Raton, NM, 1985.

Recognition and Management of Diabetic Emergencies: Hypoglycemia and Ketoacidosis, Intensive care unit nurses, St. Vincent Hospital.

What's New In Diabetes?, Entire medical staff, Socorro General Hospital, Socorro, NM, 1986.

A Comprehensive Review of Diabetes, New Mexico State Pharmacists Association (annual statewide meeting), Santa Fe, NM

Management and Differential Diagnosis of Severe Hypercalcemia, Entire medical staff, Shiprock Regional Hospital, Shiprock, NM, 1987.

Comprehensive Review of Diabetes: Physiology and Management, Entire medical staff, Health Centers of Northern New Mexico, 1987.

Diabetes Update: Theory and Practice, Statewide Family Practice Symposium, New Mexico Academy of Physician Assistants, Taos, NM, 1988.

A Practical Approach to Diabetes, (full day workshop in Diabetes), Entire medical staff, Eastern New Mexico Medical Center, Roswell, NM, 1988.

Management and Differential Diagnosis of Severe Hypercalcemia, Entire medical staff, San Juan Regional Medical Center, Farmington, NM, 1988.

Case Studies in Thyroid Disease, Entire medical staff, Holy Cross Hospital, Taos, NM, 1989.

Diabetes Update 1989, (full day workshop), Entire medical staff, Gila Regional Medical Center, Silver City, NM, 1989.

Case Studies in Thyroid Disease, Entire medical staff, Northeast Regional Medical Center, Las Vegas, NM, 1989.

Osteoporosis: Appropriate Use and Interpretation of Dual X-ray Based Absorption Densitometer, Radiology staff, St. Vincent Hospital, 1989.

Osteoporosis: Latest Methods in Diagnosis and Treatment, Entire medical staff, St. Vincent Hospital, 1989.

Osteoporosis: Etidronate, Internal Medicine Section, St. Vincent Hospital, 1990

Hypercalcemia: Case Study, Differential Diagnosis, Entire medical staff, Holy Cross Hospital, Taos, NM, 1993.

Humoral Hypercalcemia of Malignancy, Entire medical staff, Los Alamos Medical Center, Los Alamos, NM, 1993.

Demonstration of Current Reflectance Meters for Self-Blood Glucose Monitoring, Entire medical staff, St. Vincent Hospital, Santa Fe, NM, March 1993.

Diabetes Control and Complications Trial: Glucose Control Makes a Difference, Entire medical staff, St. Vincent Hospital, Santa Fe, NM, December 1994

Diabetes Control and Complications Trial: Glucose Control Makes a Difference, Entire medical staff, Los Alamos Medical Center, Los Alamos, NM, February 1995

Diabetes Control and Complications Trial: Glucose Control Makes a Difference, Entire medical staff, Holy Cross Hospital, Taos, NM, April 1995.

The Role of the Primary Care Physician in Managing Diabetic Patients, "Outcomes and Disease Management of Diabetes," (sponsored by The Zitter Group/Center for Outcomes Information), San Francisco, CA, July 1995.

Diabetic Ketoacidosis: Pathophysiology and Therapy Made Easy, "Practical Diabetes '95 Seminar", Family Practice Physicians, Hilton of Santa Fe, Santa Fe, New Mexico, October 1995.

Diabetes Control and Complications Trial (DCCT): Results and Implications for Therapy, "Practical Diabetes '95 Seminar", Family Practice Physicians, Hilton of Santa Fe, Santa Fe, New Mexico, October 1995.

Insulin Management: Easy Methods for Optimal Glucose Control, "Practical Diabetes '95 Seminar", Family Practice Physicians, Hilton of Santa Fe, Santa Fe, New Mexico, October 1995.

Diabetes: Other Than Glucose Control, What Else Is Important? "Practical Diabetes '95 Seminar", Family Practice Physicians, Hilton of Santa Fe, Santa Fe, New Mexico, October 1995.

Therapy of Type 2 Diabetes, "Practical Diabetes '95 Seminar", Family Practice Physicians, Hilton of Santa Fe, Santa Fe, New Mexico, October 1995.

Syndrome X: Insulin Resistance, Hypertriglyceridemia, Hypertension, and Coronary Artery Disease, "Practical Diabetes '95 Seminar", Family Practice Physicians, Hilton of Santa Fe, Santa Fe, New Mexico, October 1995.

Future Developments in the Therapy of Diabetes, "Practical Diabetes '95 Seminar", Family Practice Physicians, Hilton of Santa Fe, Santa Fe, New Mexico, October 1995.

Type 2 Diabetes: Appropriate Use of the New Oral Agents, General Internists and Family Practice Physicians (Statewide) Eldorado Hotel, Santa Fe, New Mexico, February 1996.

Clinical Application of Hemoglobin A1C Assay, Public Health Service Laboratory Technicians, Albuquerque, New Mexico, July 1997.

Type 2 Diabetes: Use of the New Oral Agents, American Association of Family Practice Physicians, NM State Conference, Ruidoso, New Mexico, July 1997.

Type 2 Diabetes: Overview of Appropriate and Comprehensive Care, American Association of Clinical Endocrinologists Primary Care Symposium, Albuquerque, New Mexico, November 1997.

Type 2 Diabetes: Targeting Macrovascular Risk Factors, Joslin Clinic Symposium, Santa Fe, New Mexico, March 1998.

Clinical Update, Type 2 Diabetes, General Internists and Family Practice Physicians, Los Alamos Medical Center, Los Alamos, New Mexico, April 1998.

Current Standards for Care for Type 2 Diabetes, General Internists and Family Practice Physicians, Santa Fe Public Health Service Hospital, Santa Fe, New Mexico, May 1998.

Appropriate Use of Oral Agents in Type 2 Diabetes, General Internists and Family Practice Physicians, Santa Fe Public Health Service Hospital, Santa Fe, New Mexico, May 1998.

Case Studies: Problem-Solving in Type 2 Diabetes, General Internists and Family Practice Physicians, Santa Fe Public Health Service Hospital, Santa Fe, New Mexico, June 1998.

The Environmental Ambush: Global Warming and Human Health, Medical Staff, St. Vincent Hospital, Santa Fe, New Mexico, April 1999.

Case Studies in Endocrinology and Diabetes, New Mexico Chapter, American Association of Family Practice, Ruidoso, New Mexico, June 1999.

Diabetes: Clinical Update, Medical Staff, Holy Cross Hospital, Taos, New Mexico, October 1999.

DCCT & UKPDS: Results and Implications for Therapy, "Clinical Diabetes: 1999", General Care Physicians, Hilton of Santa Fe, Santa Fe, New Mexico, October 1999.

Oral Agents in the Care of Type 2 Diabetes: A Review, "Clinical Diabetes: 1999", General Care Physicians, Hilton of Santa Fe, Santa Fe, New Mexico, October 1999.

Cardiac Risk Factor Modification in Type 2 Diabetes, "Clinical Diabetes: 1999", General Care Physicians, Hilton of Santa Fe, Santa Fe, New Mexico, October 1999.

What's New in Weight Management, "Clinical Diabetes: 1999", General Care Physicians, Hilton of Santa Fe, Santa Fe, New Mexico, October 1999.

Type 2 Diabetes: Clinical Update and Case Studies, "Diabetes Day" (NM Chapter of the American Association of Clinical Endocrinologists), Albuquerque, New Mexico, December 1999.

The Environmental Ambush: Global Warming and Human Health, Medical Staff, Presbyterian Hospital, Albuquerque, New Mexico, December 1999.

The Environmental Ambush: Global Warming and Human Health, Medical Staff, Los Alamos Medical Center, Los Alamos, New Mexico, February 2000.

Type 2 Diabetes: Clinical Update, Los Alamos Medical Center Grand Rounds, Los Alamos, New Mexico, February 2000.

The Environmental Ambush: Global Warming and Human Health, The Research Association of Medical and Biological Organizations, Albuquerque, New Mexico, February 2000.

Type 2 Diabetes: Clinical Update, Espanola Hospital, Espanola, New Mexico, February 2000.

Global Warming and Human Health, General Medical Staff, Lovelace Medical Center, Albuquerque, New Mexico,

November 2000.

Global Warming and Human Health, General Medical Staff, Holy Cross Hospital, Taos, New Mexico, November 2000.

Diabetes 2003: Clinical Update, Grand Rounds, St. Vincent Hospital, Santa Fe, NM, March 2003.

Global Warming, Human Health, and Energy Choices, Grand Rounds, Santa Fe Public Health Hospital, Santa Fe, NM, November 2003.

Global Warming and Human Health, NM Council of Churches, Global Warming Conference, Albuquerque, NM, November 2003.

Diabetes Update: 2004, Grand Rounds, St. Vincent Hospital, Santa Fe, NM, April 2004.

Thyroid Cancer, Grand Rounds, St. Vincent Hospital, Santa Fe, NM, November 2004.

Osteoporosis, Grand Rounds, St. Vincent Hospital, Santa Fe, NM, February 2005.

Type 2 Diabetes, Newer Therapies, Grand Rounds, Holy Cross Hospital, Taos, NM, October 2005.

Case Studies in Endocrinology, Grand Rounds, St. Vincent Hospital, Santa Fe, NM, June 2007.

Global Warming and Human Health, Gallup Solar Group, Gallup, NM, June 2007.

Insulin Pump Safety, Diabetes Technology Society, Bethesda, MD, November 2008.

Insulin Pumps and Continuous Glucose Monitoring Systems, Grand Rounds, St. Vincent Hospital, Santa Fe, NM, January 2010.

Immunomodulation in Type 1 Diabetes, General Medical Staff of St. Vincent Hospital, Santa Fe, NM, May 2011.

Autoimmunity in Type 1 Diabetes, Grand Rounds, St. Vincent Hospital, Santa Fe, NM, June 2011.

Diabetes and Prediabetes, American Association of Clinical Endocrinologists, Endocrinology Day, Albuquerque, NM, March 2012.

Case Reports: Problems with Insulin Pumps, FDA Meeting on Pump Technology, Bethesda, MD, January 2013.

Diabetes and Cardiovascular Disease, American Association of Clinical Endocrinologists, Diabetes Day, Albuquerque, New Mexico, February 2015.

Diabetes Update, La Familia Medical Staff, Santa Fe, NM, March 2016.

Diabetes: New Therapies, Grand Rounds, St. Vincent Hospital, Santa Fe, NM, August 2017.

Thyroid Nodules, La Familia Medical Staff, Santa Fe, NM, September 2017.

Health Impacts of Fossil Fuels, New Mexico Public Health Association, Santa Fe, NM, December 2018.

Uncontrolled Type 2 Diabetes, La Familia Medical Staff, Santa Fe, NM, May 2021.

CLINICAL PRACTICE GUIDELINES

Member of expert panel for development of diabetes care guidelines, Value Health Sciences, Santa Monica, California.

Member of expert panel for development of diabetes care guidelines, Rocky Mountain HMO, Denver, Colorado.

Featured speaker, and member of expert panel for development of diabetes care guidelines, Integrated Pharmaceutical Services/Foundation Health Plan, Santa Monica, CA.

The above three positions included extensive participation in development of care guidelines for Type 2 diabetes, including guidelines for initiating and adjusting oral medications, insulin, hypertension and lipid control, and other aspects of comprehensive diabetes care.

American Association of Clinical Endocrinologists: Member of Committee on Clinical Practice, Standards and Guidelines 1998.

American Association of Clinical Endocrinologists: Ad Hoc Committee on Medicare Diabetes Mellitus Treatment Provisions, 2001.